# AI-Powered Intelligent Job Search and Career Autopilot

## Data Science and AI Project

### Milestone 1 Report

**Group 4**

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## 1. Problem Statement, Stakeholders and Scope

The problem addressed in this project is the lack of an integrated, intelligent, and candidate-specific system for managing the modern job search workflow. A candidate today typically has to upload or rewrite resumes, search multiple job portals, compare job descriptions manually, guess which roles fit their background, tailor resumes for each role, check ATS compatibility, and write cover letters separately. These steps are fragmented across different tools, and the candidate becomes the “manual integration layer” between them.

The proposed system, AI-Powered Intelligent Job Search and Career Autopilot, aims to solve this by building a multi-agent web application that converts an uploaded resume into a structured candidate profile, predicts suitable career roles, discovers relevant job opportunities, ranks them by fit, and generates role-specific tailored resumes and cover letters grounded in the candidate’s actual profile.
The primary stakeholders are active job seekers, fresh graduates, career changers, and working professionals who want to apply to better-matched roles with less repetitive manual effort. The secondary stakeholders are recruiters and employers, who may benefit indirectly from receiving more relevant, better-structured, and role-aligned applications.

The project is in scope for the following:

| Area | Included in Scope |
|---|---|
| Resume ingestion | Parsing uploaded resumes and extracting structured fields such as skills, education, experience, projects, certifications, and job titles |
| Candidate profile creation | Building a persistent structured profile that later agents can use |
| Career role prediction | Recommending suitable roles and career directions based on the candidate’s profile |
| Job discovery and ranking | Searching relevant job postings and ranking them by match score |
| Resume tailoring | Modifying resume content for a specific job description while preserving factual correctness |
| ATS scoring | Providing an ATS-style compatibility score based on skills, keywords, role fit, and formatting |
| Cover letter generation | Creating role-specific cover letters grounded in the candidate profile and job description |
| Web interface | A usable application interface where candidates can upload resumes, view recommendations, and generate application material |

The project is **explicitly out of scope** for the following:

- Checking the authenticity or genuineness of posted jobs, including background verification of employers or companies.
- Fully autonomous auto-apply without user review.
- Extracting personal contact information of recruiters
- Direct messaging to recruiters for users.
- Interview preparation
- Fabricating experience or credentials

## 2. Why the Problem Matters and Where It Arises

This problem matters because job search has become increasingly competitive, AI-mediated, and time-consuming. LinkedIn's 2026 research reports that more than half of people globally are looking for a new role and 65% of people say finding a job has become more challenging. The same report also states that 93% of recruiters plan to increase their use of AI in 2026, showing that both sides of the hiring process are becoming more AI-driven.

This problem arises in online recruitment settings where job seekers use platforms such as LinkedIn, Naukri, Indeed, Glassdoor, company career pages, and ATS resume-checking tools. However, the existence of many portals does not automatically make the process easier. Candidates still face three practical difficulties:

- **Discovery overload:** There are many listings, but the candidate does not know which ones truly match their profile.
- **Application repetition:** Every serious application requires resume tailoring, keyword alignment, and a role-specific cover letter.
- **Lack of feedback:** Candidates rarely know whether their resume is aligned with a job description before applying.

Therefore, the project is important because it targets the gap between job availability and candidate readiness. It does not simply search for more jobs; it helps the candidate identify better-fit jobs and prepare higher-quality application material for each one.

## 3. How the Problem Is Currently Addressed

At present, the job search workflow is supported by both commercial tools and academic research. Commercial tools usually handle one stage of the process, while academic research focuses on individual technical tasks such as resume parsing, job matching, career recommendation, and text generation.

### a. Commercial tools

- **Job discovery**: It is mainly handled by platforms such as LinkedIn, Naukri, Indeed, Glassdoor, and company career pages. These platforms provide search filters, saved alerts, and job recommendations. However, the candidate still has to manually inspect each job description and decide whether it is a good fit.
- **ATS scoring and optimization**. These are handled by tools such as Jobscan and SkillSyncer. These tools compare a resume with a job description and identify

missing keywords or skills. However, they are separate from job discovery platforms, so the user has to copy and paste job descriptions manually.

- **Resume and cover letter writing:** It is commonly done using resume builders or general-purpose LLMs. These tools can generate fluent text, but if they are not connected to a verified candidate profile, the output may become generic or may include unsupported claims.
- **Career guidance:** It is often provided through static guides, skill taxonomies, or simple recommendation tools. These suggestions are usually role-level and generic, instead of being based on what the individual candidate has actually worked on.

### b. Academic Approaches

- **Resume Parsing:** Resume parsing is treated as an information extraction task. Transformer models like BERT are useful for extracting skills, education, experience, and job titles from text. For PDF or scanned resumes, layout-aware models like LayoutLM can also use document structure, not just text.
- **Resume Classification and Profile Understanding:** Recent work such as ResumeAtlas shows that large resume datasets and LLM-based methods can help classify resumes and understand candidate profiles. It also highlights challenges such as privacy, different resume formats, and data quality.
- **Job Matching:** Traditional job matching uses keyword overlap, but this can miss similar skills or related meanings. Semantic embedding models like Sentence-BERT compare resumes and job descriptions by meaning. Other person-job fit models, such as Qin et al. and ConFit, use neural networks and contrastive learning to improve resume-job matching.
- **Career Recommendation:** Career recommendation can use recommender systems, knowledge graphs, and skill-role taxonomies. Methods such as Dave et al. learn relationships between jobs and skills. Public taxonomies like O*NET and ESCO help map skills, occupations, and career paths.
- **LLM-Based Resume and Cover Letter Generation:** LLMs can generate resumes and cover letters, but they may create unsupported claims if not grounded. RAG-based methods reduce this issue by giving the model verified context. In this project, the LLM will use only the candidate profile and job description to generate grounded outputs.

## 4. Common Metrics and Existing Definitions of Success

Existing work uses different metrics depending on the stage of the workflow.

### Metric Group 1: Quantitative Metrics

**a. Precision, Recall, and F1-Score**

Applied to **resume extraction**. These metrics check how accurately the system extracts structured details such as skills, education, experience, projects, and job titles from a resume. The extracted results can be compared with manually checked ground truth data.

**b. Top-K Accuracy and Mean Reciprocal Rank (MRR)**

Applied to **career path prediction**. Top-K Accuracy checks whether the correct or suitable career role appears in the top suggested roles. MRR checks how high the first correct recommendation appears in the ranked list.

### Metric Group 2: Qualitative Metrics

a. **LLM-as-Judge Scoring**
Applied to **resume tailoring, cover letter generation, and career recommendations**. A separate language model can rate the output on a 1 to 5 scale using criteria such as relevance, clarity, personalization, factual accuracy, and role fit.

b. **Human Review**
Human reviewers can check whether the tailored resume and cover letter are specific, professional, and based only on the candidate's real profile. This is important because generated text should not include fake skills or unsupported claims.

### How Success is Defined in Existing Systems

Academic systems measure success using proxy evaluation metrics such as F1-score for resume parsing and Top-k Accuracy/MR for career prediction, as real hiring outcomes are typically unavailable. Commercial platforms evaluate success using user and business outcomes, including application conversion, interview callback rates, and time-to-hire. For this project, success is defined by strong performance on these established evaluation metrics together with qualitative assessment of generated resume and cover letter quality.

## 5. Where Current Solutions Fall Short and Opportunity for This Work

The main problem with current job search tools is that they are disconnected. Candidates use one tool to search for jobs, another to build resumes, another to check ATS score, and another to write cover letters. These tools do not work together, so the candidate has to do most of the work manually.

This creates the following gaps that our project can address:

- **Job discovery is limited to major job boards.**
Most candidates search only on popular platforms like LinkedIn, Naukri, Indeed, and Glassdoor. These platforms show highly visible jobs, but those jobs also attract many applicants. Many useful openings on company career pages or smaller job boards may be missed. Our project can use a job discovery agent to search more widely and find roles that better match the candidate's actual profile.
- **The workflow is fragmented.**
Resume writing, job search, ATS checking, and cover letter generation are usually handled by separate tools. The candidate has to copy and paste information between them. Our project solves this by using one shared candidate profile across all agents, so the process becomes more connected and easier to use.
- **Resume and cover letter generation is often generic.**
Many AI tools can write polished text, but the output may not be specific to the candidate's real skills, projects, and experience. This can make the resume or cover letter look generic. Our system will generate content based on the verified candidate profile and the target job description, making the output more personal and relevant.
- **There is no strong feedback loop.**
Current tools may generate a resume or cover letter, but they do not always show how well it matches the job. Our project adds ATS-style scoring so the system can improve the resume based on the score and job requirements.

- **Career guidance is usually static.**
  Many career recommendation tools give general suggestions and do not update as the candidate learns new skills or adds new projects. Our system can update recommendations when the candidate profile changes, helping the user find better-fitting roles over time.

Overall, the opportunity for our project is to build a connected, profile-based, multi-agent system that supports the candidate from job discovery to tailored application material in one workflow.

## 6. Nature of the Project Contribution

The contribution of this project is not a claim of beating academic benchmarks. It is positioned as a practical system contribution across following dimensions:

- **Integration and usability:** The project combines resume parsing, career prediction, job ranking, ATS scoring, resume tailoring, and cover letter generation into one connected pipeline. This reduces the need for candidates to manually move between separate tools.
- **Grounded generation:** The system will generate resumes and cover letters only from verified candidate profile data and the target job description. This is a design constraint to reduce hallucinated achievements, fake skills, and generic content.
- **Quality-first application support:** Instead of encouraging blind auto-apply behaviour, the system emphasizes fit scoring, explainable ranking, ATS improvement, and human review before submission.

Therefore, the main contribution lies in usability, integration, grounded generation, and the practical use of multiple AI agents in one job-search workflow.

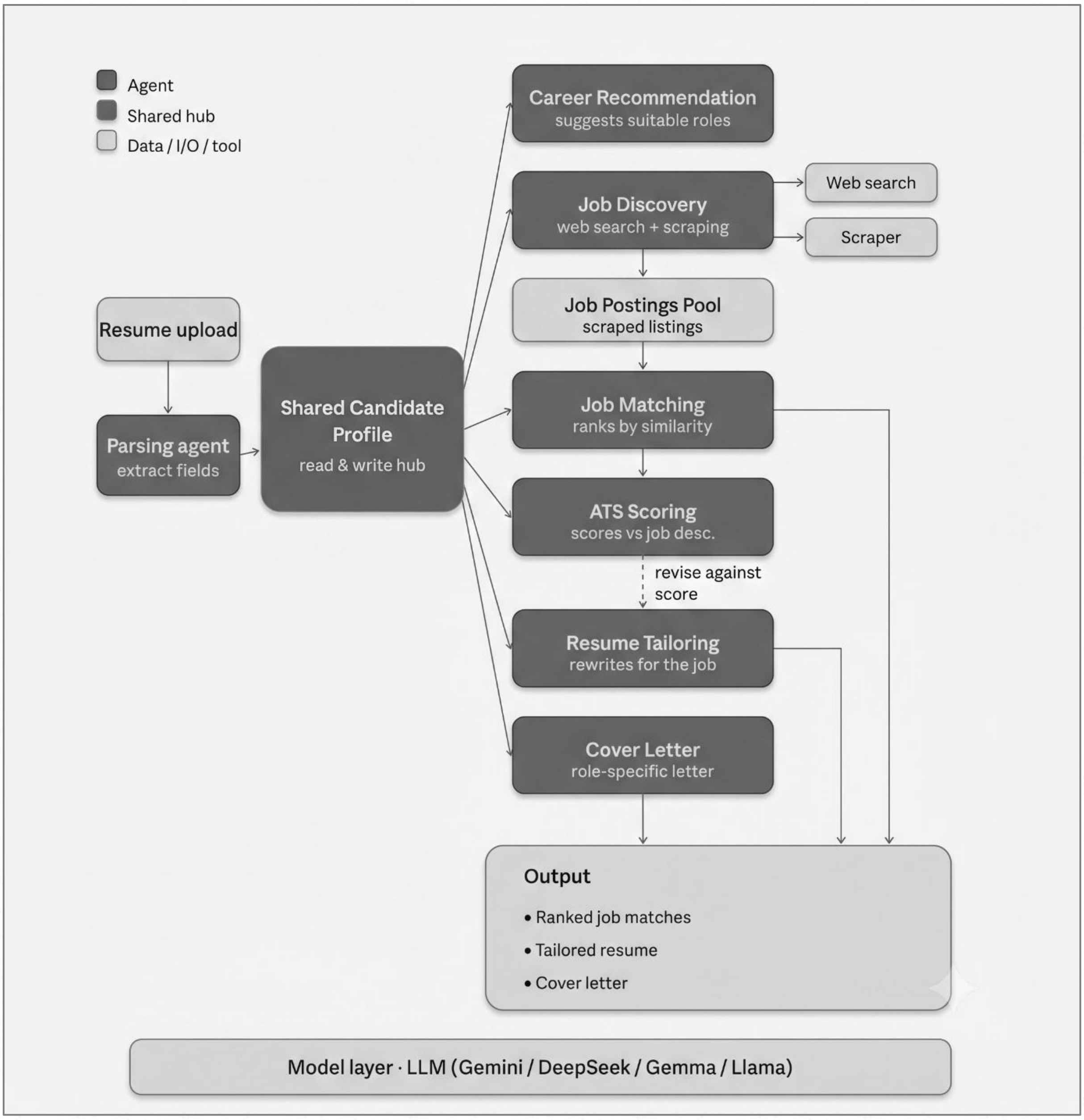
## 7. Proposed System Architecture / Workflow

The system uses a **multi-agent AI pipeline** built around one **shared candidate profile**. The user uploads a resume, and the **Parsing Agent** extracts skills, education, experience, projects, and job titles into a structured profile.

The **Career Recommendation Agent** suggests suitable roles from this profile.
The **Job Discovery Agent** searches public job sources and creates a pool of job postings.

The **Job Matching Agent** ranks jobs by similarity with the candidate profile.
The **ATS Scoring Agent** checks how well the resume matches a selected job description.

Finally, the **Resume Tailoring Agent** and **Cover Letter Agent** generate role-specific application material.
All agents use the LLM model layer, with models such as **Gemini, DeepSeek, Gemma, or Llama** under consideration.




## 8. Preliminary Dataset Plan and Challenges

The project does not depend on a single dataset. Instead, different data sources will be used for different parts of the pipeline.

| Pipeline Component | Expected Data Source | Purpose |
|---|---|---|
| Resume parsing | Public Kaggle resume datasets, open-source resume samples. | To extract skills, education, experience, projects, certifications, and job titles |
| Candidate profile creation | Parsed resume data | To create a structured candidate profile used by all agents |

| | | |
|---|---|---|
| Job discovery and matching | Public job description datasets, company career pages, and India-focused job aggregators where allowed | To collect job title, company, location, required skills, experience, job type, and description |
| Career Recommendation | O*NET, ESCO, and public skill-role taxonomies | To map candidate skills and experience to suitable job roles |
| Evaluation data | Small, curated set of resumes, job descriptions, expected role suggestions, tailored resumes, and cover letters | To evaluate output quality and system usefulness |

### Anticipated Challenges

**Data availability:**
There may not be one complete dataset that contains resumes, job descriptions, career roles, ATS scores, and cover letters together. Therefore, the project will combine multiple smaller datasets and manually curated examples.

**Privacy:**
Real resumes may contain personal information. Any volunteer resumes used in the project should be anonymized before processing.

**Dynamic job postings:**
Online job posts change frequently. Some postings may expire or be removed. For the prototype, saved job descriptions or public datasets can be used to keep evaluation stable.

## 9. Evidence That Will Demonstrate the Approach Works

At this stage, the aim is not to prove the system with large-scale testing, but to show that the proposed Generative AI pipeline works in a clear and practical way.

| **Component** | **Evidence of Success** |
|---|---|
| Resume Parsing | The system should correctly extract important details such as skills, education, experience, projects, and job titles from an uploaded resume. |
| Candidate Profile Creation | The extracted resume information should be converted into a clear structured profile that can be used by other agents. |
| Career Role Prediction | The system should suggest suitable job roles based on the candidate's skills, education, and experience. The suggestions should be reasonable and explainable. |
| Job Matching | The system should rank jobs based on how well they match the candidate profile. Better-fitting jobs should appear higher in the list. |

| | |
|---|---|
| ATS Score Improvement | The system should compare the original and tailored resume for a selected job description and show improvement in ATS-style score. |
| Resume Tailoring | The tailored resume should be specific to the selected job role and should not add fake skills or false experience. |
| Cover Letter Generation | The generated cover letter should be personalized, role-specific, and based on the candidate's actual profile. |
| Grounded Generation | The system should only use information available in the resume/profile and job description. Any unsupported or invented claim will be treated as an error. |
| User Experience | A user should be able to complete the flow from resume upload to job match, resume tailoring, and cover letter generation through the web interface. |

Overall, the project will be considered successful if it shows that a multi-agent pipeline can reduce manual effort in the job search process and produce better, more personalized application material than using separate tools manually.